Introduction to Population Biology

Second Edition

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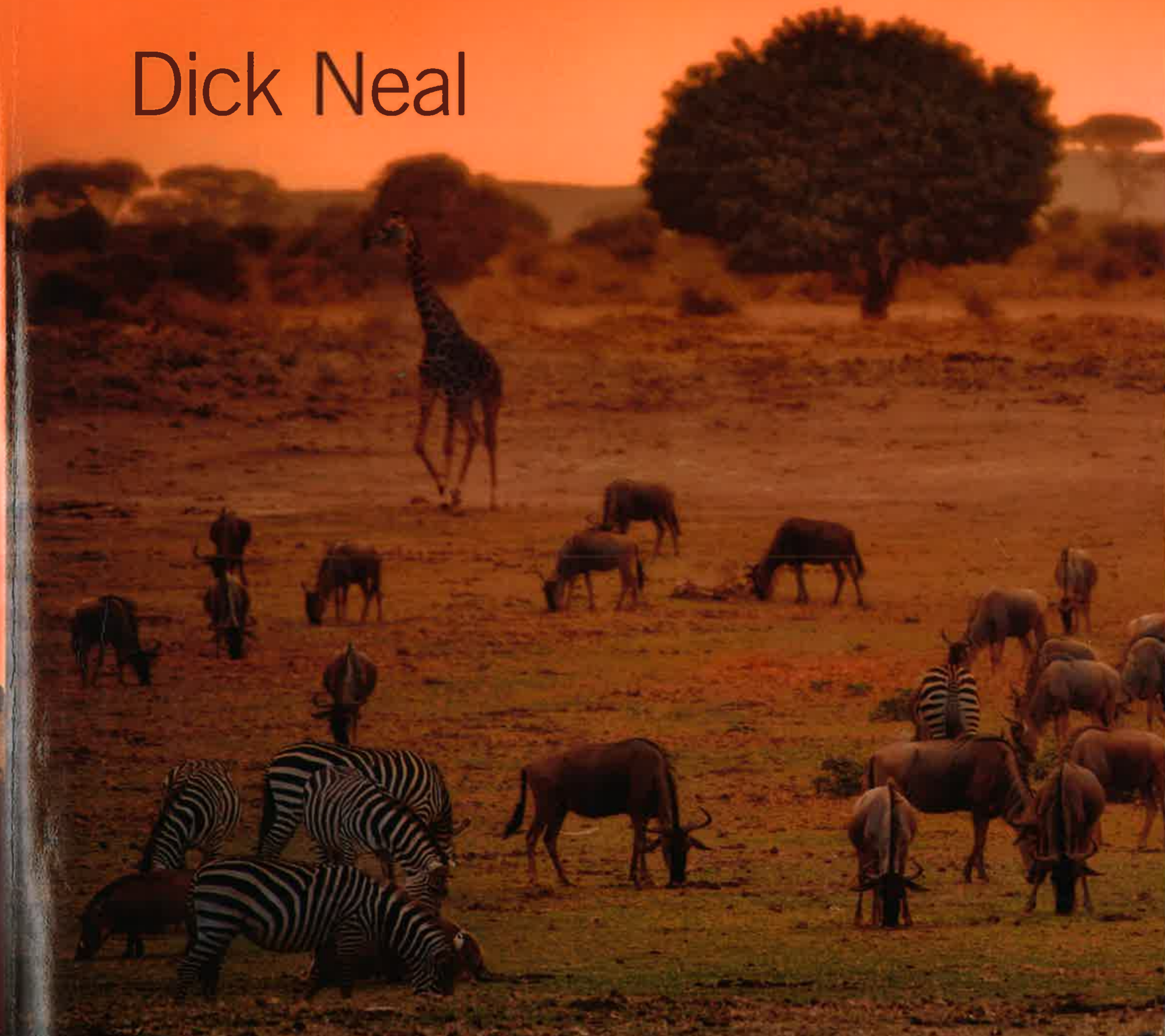

6 Life Tables

This chapter considers how survival is influenced by an individual's age and sex, and reviews how information on age-specific death rates is tabulated in an orderly way in the form of life tables. In some animals, such as insects, this classical development of life tables is frequently modified to consider life history stages, rather than chronological age, because the duration of a stage may vary considerably in relation to environmental conditions. Although life tables have been developed for some plants (see Harper 1977), their phenotypic plasticity often makes the life table approach inappropriate. For example, some stages (seeds and tubers) may lie dormant for years before starting or resuming active growth, and other stages may revert back to an earlier life history stage after part of an individual has been eaten by herbivores or killed by fire. We will consider how to resolve such difficulties using matrix models in Chapter 7.

6.1 DEVELOPING A LIFE TABLE

Survival rates, or, conversely, death rates, vary with both the sex and age of an individual, and are also affected by the environment. We keep track of these statistics in a very orderly way, but demographers are optimistic people and so rather than talking of death tables they develop *life tables*, and use survivorship data rather than age-specific death rates in growth models. To see how life tables are developed, we will initially consider the hypothetical example in Table 6.1. This represents the fate of a cohort of individuals of one sex born at one time, typically a breeding season, although it can also represent several cohorts to give an average over a series of breeding seasons. This may be necessary if the sample size in a single breeding season is small.

1. The first column is an age interval (x), the duration of which will depend on the organism being studied. Usually, the time intervals are of equal duration, but occasionally some ages are grouped together and the interval may be indicated as, for example, <5 years. Frequently, the age interval is only indicated by the start of the interval, i.e. 0, 1, 2, 3, etc.
2. The second column shows the number of survivors (S_x) at the start of the age interval. Sometimes, it is practically impossible to sex individuals at this stage. In this case, we may assume that half are male and half female, and in this case we usually put the starting number in parentheses. Their survivorship is followed, in this example year by year, until they had all died by the end of their sixth year of age. Note that it is common for there to be some

Table 6.1 **Hypothetical cohort life table to illustrate how the various parameters are calculated (see text)**

Age interval (years)	Number at start of interval	Number dying in interval	Probability of survival from birth	Fraction dying in interval	Probability of dying in interval	Mean expectation of life
x	S_x	D_x	l_x	d_x	q_x	e_x
0–1	100	60	1.00	0.60	0.600	1.85
1–2	40	4	0.40	0.04	0.100	2.88
2–3	36	9	0.36	0.09	0.250	2.14
3–4	27	9	0.27	0.09	0.333	1.69
4–5	18	8	0.18	0.08	0.444	1.28
5–6	10	6	0.10	0.06	0.600	0.90
6–7	4	4	0.04	0.04	1.000	0.50
7–8	0	–	0	–	–	–
		$\sum = 100$		$\sum = 1.00$		

uncertainty about the number of individuals in the first age class. For example, we may estimate the number of eggs laid by a fish by counting the number of eggs in adult females and assuming that all are laid.

3. The third column shows the number of individuals dying (D_x) within each age class. This is easy to compute if the number of survivors at the start of each age class is known. The number dying in the first age class is 100 − 40 = 60; the number dying in the second age class is 40 − 36 = 4, and so on. Symbolically, this is calculated as $D_x = S_x - S_{x+1}$. We can check our arithmetic by summing the total dying in all age classes. This should equal the number alive at the start of the first age class.
4. To make it easier to compare life tables with very different starting numbers, the number of survivors is converted to a probability of survival series, l_x, by dividing the number of survivors, S_x, at each age by the number alive at the start of the first age class. Thus, $l_0 = 100/100$ or 1.0, $l_1 = 40/100$ or 0.4, $l_2 = 36/100$ or 0.36, and so on. Often series of $100l_x$, $1000l_x$, $10\,000l_x$, and so on are calculated so that one starts with 100, 1000 or more individuals and the following number of survivors will appear as integers rather than decimals.
5. The d_x series is calculated in exactly the same way as the D_x series (column 3) by using the data in column 4. Symbolically, this is calculated as $d_x = l_x - l_{x+1}$, and so $d_0 = 1.0 - 0.4$ or 0.6, $d_1 = 0.4 - 0.36$ or 0.04, and so on. It is easy to introduce rounding errors here, and it is better to use the same conversion rate for D_x to d_x as was used for S_x to l_x.

6. The sixth column computes the probability of dying, or the age-specific mortality rate (q_x), during each age interval. This is calculated by either D_x/S_x or d_x/l_x. It is better to use D_x/S_x to avoid rounding errors. Thus, q_0 = 60/100 or 0.6, q_1 = 4/40 or 0.1, and so on.
7. Finally, the mean life expectancy (e_x) is calculated for different ages in the last column. This is estimated by summing the number of survivors at the start of the age class (x) and older age classes, and dividing this sum by the number of survivors in the age class (x). This estimate assumes that all individuals alive at the start of an age class survive to the end of that age class and then die. *If all age classes are of equal duration*, we can correct for this by subtracting half of the value of the age interval (x) from our estimate, assuming that mortality occurs evenly throughout each age interval. Symbolically, this may be stated as:

$$e_x = [(S_x + S_{x+1} + S_{x+2} \ldots S_n)/S_x] - 0.5x \qquad \text{(Eqn 6.1)}$$

where S_n is the number in the oldest age class. Thus, from Table 6.1 E_0 = [(100 + 40 + 36 + 27 + 18 + 10 + 4)/100] – 0.5, or 1.85 years, E_1 = [(40 + 36 + 27 + 18 + 10 + 4)/40] – 0.5, or 2.875 years, E_2 = [(36 + 27 + 18 + 10 + 4)/36] – 0.5, or 2.139 years, and so on. This method gives the same values of e_x provided by the more usual equation that is used if the age classes are not of equal duration:

$$e_x = \frac{1}{l_x}\sum_{y-x}^{n}\frac{(l_y + l_{y+1})}{2} \qquad \text{(Eqn 6.2)}$$

I suggest that the S_x values be used rather than the l_x values because the latter can introduce significant rounding errors. It may surprise you that the mean expectation of life can increase with age for part of the life table. This happens following a sharp decrease in mortality rates.

6.2 DIFFERENT TYPES OF LIFE TABLES

We will now look at the different ways of obtaining information on age-specific mortality and show how different types of life tables are developed. Some methods have rather restrictive assumptions, and the different types of life tables are applied to different situations. The original data and analyses for all of the life tables in this chapter are provided in the Simulations website for Chapter 6.

6.2.1 Cohort (Horizontal or Generation) Life Tables

In this type of life table, the survival of a known group of organisms, called a cohort, is followed from birth to the time when they are all dead. Cohort life tables can be applied to populations at any phase of population growth, i.e. the populations could be increasing or decreasing, or be stationary in size. The method is typically applied to organisms that can be marked soon after birth and their survival followed individually.

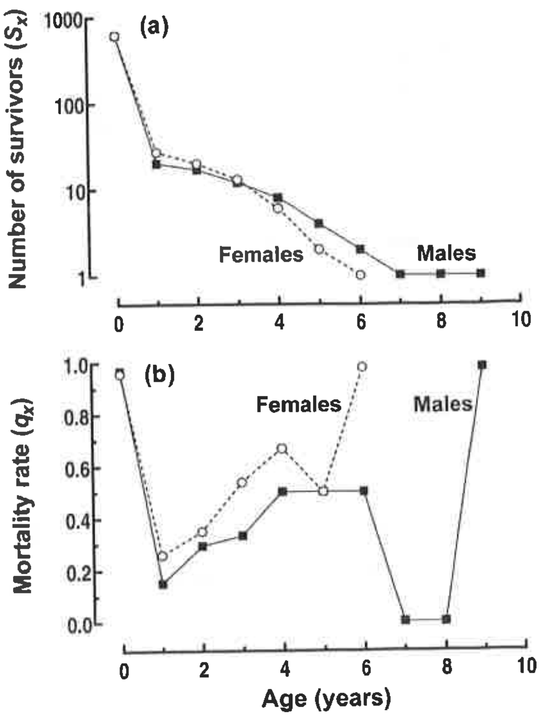

Figure 6.1 (a) Survivorship of male and female *Geospiza conirostris* that were banded as nestlings during the period 1978-83.
(b) Age-specific mortality rates of the two sexes. (Data from Grant and Grant, 1989.)

An example of a cohort life table, or more correctly a composite cohort life table, was developed for the large cactus (ground) finch, *Geospiza conirostris*, on Genovesa Island in the Galápagos archipelago by Peter and Rosemary Grant (Grant and Grant 1989). The survival and age-specific mortality rates of males and females are illustrated in Figure 6.1.

The shape of the survivorship curve was similar for the two sexes (Figure 6.1a). After the nestling stage, males generally survived slightly better than females, with some males surviving for more than 9 years, whereas all females were all dead by 7 years of age. Note that the number of survivors is plotted on a logarithmic scale. If they had been plotted arithmetically, the shape of most of the curve would have been obscured because more than 95 per cent of their mortality occurred during the first year of life. A logarithmic plot also has an added advantage because the slope of the graph gives us a measure of the age-specific death rates, i.e. the slope of the curve is directly related to the mortality rate. Thus, we can see that the mortality rate is highest during the first year of life, declines to its lowest rate between 1 and 2 years of life, and thereafter generally increases with age.

The age-specific mortality rates of the two sexes follow a U-shaped curve (Figure 6.1b), with males generally having lower mortality rates than females, except at the nestling stage.

The Grants collected and combined their data for the period 1978–83. They could not sex the nestlings and so, of the 1244 nestlings they marked, they made the reasonable assumption that half (622) were male and half were female. They combined the data from several years into a composite cohort life table because the very heavy early mortality (>95 per cent) meant that the numbers of older birds were low; in total, only 27 females and 20 males survived for 1 year. Nevertheless, the Grant's were able to show that the survival of the adults of different cohorts systematically changed by following the median survival age (the age at which 50 per cent of the birds starting their second year of life had died). For males, the median survival age was reached in the sixth year by the 1976 cohort, in the fifth year by the 1978 cohort, in the fourth year by the 1979 cohort, in the third year by the 1980 cohort, and in the second year by the 1981 and 1982 cohorts. The female median survival age was usually 1 year less than that of the males.

How is this pattern of mortality over time explained? Adult mortality occurs mainly during the non-breeding dry season when food is scarcer and the birds spend much of their time searching for food. The period between 1978 and 1983 was favourable for these birds, and the population approximately quadrupled during this time, increasing intraspecific competition for food. It appears that the later cohorts were progressively less able to compete against and displace the adults of earlier cohorts, and so suffered increasing mortality.

Clearly, survival is not random with respect to sex, age, year of birth, various environmental conditions, and so on.

6.2.2 Time-Specific or Vertical Life Tables

This method can be used for any species but is particularly useful for long-lived species where it is generally impractical to develop a cohort life table. At a single time, each age class in the population is studied to estimate their age-specific death rates independently over the same time period (e.g. 1 year), hence the time-specific label. The estimates are then combined to develop the full life table for that particular time in the following manner.

Imagine following N_x individuals in each age class of a population and determining their fate over 1 year (Table 6.2). We can see that five of 200 newborns died during their first year ($D_x = 5$), giving an estimated annual mortality rate ($q_x = D_x/N_x$) of 0.025 for the age class 0–1 years. In the second age class (2–5 years), we see that only one individual died in the same year (D_x in column 3) out of a sample of 200, so the annual mortality rate (q_x) is 1/200 = 0.005. In the same way, we can complete the first five columns of Table 6.2.

Using this information we can now construct a life table (columns 6–8) for this population. First, we calculate columns 6 and 7 in an alternating manner. For the first age class, the starting value in column 6 is 1000, and we see that column 7 equals the number of years in the age class (column 2) multiplied by the annual death rate in column 5, multiplied by the number of individuals at risk (the starting number in column 6). Thus $1000d_x$ is $1 \times 0.025 \times 1000 = 25$. We can then calculate the starting number in the second age class of column 6 as $1000 - 25 = 975$, and the value for column 7 is $4 \times 0.005 \times 975 = 19.5$. Therefore, the value of column 6 for the third age class is $975 - 19.5 - 955.5$. In this manner, we can complete columns 6 and 7. The $1000q_x$ values in column 8 are easy to calculate because they are simply 1000 times the value of q_x in column 5. For further information, see the Simulations website for Chapter 6, vertical life tables.

Table 6.2 **An artificial example to illustrate how a time-specific life table is developed. Samples of individuals (N_x) of each age class are followed over a specific time period (here 1 year) to obtain age-specific mortality rates (q_x). The vertical life table can then be calculated (see text and Simulation website for details)**

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Age class (years)	Number of years in age class	Sample size	Number dying in 1 year	Annual death rate (D_x/N_x)	Survivors at start of age class	Deaths in age class: (2) × (5) × (6)	Annual death rate: (5) × 1000
(x)		(N_x)	(D_x)	(q_x)	($1000l_x$)	($1000d_x$)	($1000q_x$)
0–1	1	200	5	0.025	1000	25	25
2–5	4	200	1	0.005	975	19.5	5
6–10	5	200	3	0.015	955.5	71.66	15
11–15	5	200	4	0.02	888.84	88.38	20
16–20	5	200	10	0.05	795.45	198.86	50
21–25	5	100	15	0.15	596.59	447.44	150
26–30	5	50	10	0.2	149.15	149.15	200
31–35	–	–	–	–	0	–	–

Now that we understand how this type of life table is developed, let us consider human life tables that use this method. Human life tables for 37 countries are freely available from the Human Mortality Database, University of California, Berkeley (USA) and the Max Planck Institute for Demographic Research (Germany), and may be downloaded at www.mortality.org or www.humanmortality.de. If we look at the 1980 statistics for the Canadian population, we see that females survived better than males (Figure 6.2a) in contrast to the finch that we looked at earlier (Figure 6.1a). The heavy early mortality is obscured in Figure 6.2a but becomes evident when we look at the annual age-specific mortality rates in Figure 6.2b. The mortality rate during the first year of life is only exceeded when males reach 60 years of age and females 65 years of age. The mortality rate is lowest at about the time of puberty. Males have consistently higher mortality rates than females but the rates converge to be similar at 100 years and above.

If we compare survival and age-specific mortality rates for Canadian females from 1921 to 2007 (Figure 6.3), it can be seen that there has been a systematic improvement in survival over this period of time. The difference in age-specific mortality is particularly striking for the early age classes (Figure 6.3b). For example, the age-specific death rate during the first year of life in

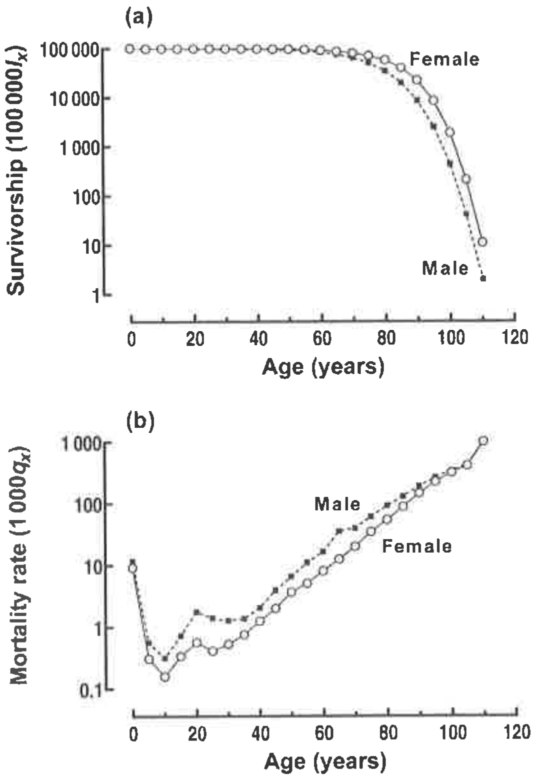

Figure 6.2 (a) Comparison of the estimated survivorship of Canadian males and females in 1980. (b) Comparison of the age-specific mortality rates of Canadian males and females in 1980. (Data from the Human Mortality Database, University of California, Berkeley, USA, available at www.mortality.org.)

2007 was 4.76 per 1000, approximately half that in 1980 (9.25 per 1000) and 1/20th that in 1921 (99.58 per 1000). Much of the improvement in early survival is related to health improvements, particularly comprehensive vaccination and inoculation to prevent childhood infectious diseases, and also clean water and better sanitation.

The Human Mortality Database provides a rich source of information about how the mortality of different populations has changed over the years. For some countries, the database covers a sufficient period for cohort life tables to be constructed.

6.2.3 Stationary or Static Life Tables

A stationary or static life table uses either the age structure of a population or the ages at which individuals die in the population to derive a life table. The method has three basic assumptions: that the population has a stationary age distribution, that the age-related patterns of births and deaths are constant, and that the number of births balances the number of deaths, i.e. $r = 0$ or $\lambda = 1$. If the population is increasing or decreasing exponentially and has attained a stable age distribution, it is possible to derive the stationary age distribution (Caughley 1977), but these conditions seem unlikely to be true for most long-lived populations. In addition to the three basic assumptions, there are technical questions such as the ability to obtain unbiased samples of the population, and the accuracy of aging living or dead individuals. In the light of such

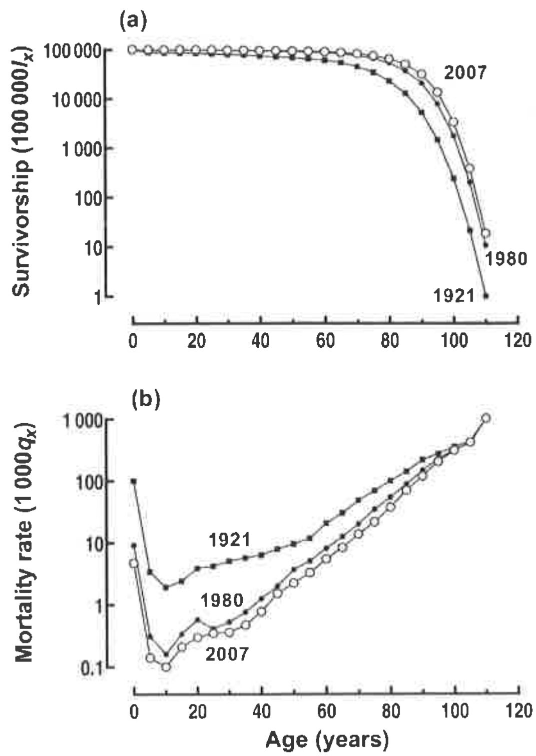

Figure 6.3 Comparison between (a) years of survivorship and (b) age-specific mortality rates of Canadian females in 1921, 1980 and 2007. (Data from the Human Mortality Database, University of California, Berkeley, USA, available at www.mortality.org.)

uncertainties, it is not surprising that some advocate 'Never, ever use static life tables' (Gurevitch *et al.* 2006). It is wise to exercise caution in applying this method.

Stationary Life Tables Based on the Age Structure of the Population

When the number of births balances the number of deaths, the age structure of the population is the same as the survivorship (S_x) series. If we determine the age structure by killing a sample from the population, the ages represent the survivorship (S_x) series. There have been instances where an investigator has assumed that the age structure represents the ages at death, i.e. the D_x series, because the animals were killed by the sampling method (Caughley 1966).

In order to understand some of the difficulties of applying this method, we will consider Graeme Caughley's classic study of female Himalayan tahr (*Hemitragus jemlahicus*) in New Zealand (Caughley 1966). His study set a new standard for the development of life tables for wild mammalian populations, and showed that most of the previously developed life tables for mammals were unacceptable. The Himalayan tahr population was sampled by hunters, and animals were aged by the growth rings on their horns. He found no evidence that the sampling of females was biased with respect to age, and showed that the data were consistent with a stationary population. His arguments do not preclude the possibility that the population was fluctuating around a stable carrying capacity (K) resulting from variations in birth rates and early mortality, which would cause fluctuations in the survivorship series. Note (column 2 in Table 6.3)

Table 6.3 **Partial life table for female Himalayan tahr (*Hemitragus jemlahicus*), in New Zealand, constructed from the age structure of the population**

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Age (years)	Frequency in sample	Adjusted frequency	Female births per female	Survivorship	Deaths in age class	Mortality rate
x		S_x	m_x	$1000l_x$	$1000d_x$	$1000q_x$
0	–	203[a]	0.000	1000	511	511
1	94	99.22	0.005	489	19	39
2	97	95.35	0.135	470	37	79
3	107	87.90	0.440	433	50	115
4	68	77.75	0.420	383	58	151
5	70	65.98	0.465	325	60	185
6	47	53.72	0.425	265	58	219
7	37	41.96	0.460	207	52	251
8	35	31.44	0.485	155	44	284
9	24	22.60	0.500	111	34	306
10	16	15.59	0.500	77	26	338
11	11	10.32	0.470	51	19	373
12	6	6.55	0.470	32		
13	3	(3)	(0.350)			
14	4	(4)	(0.350)			
15	3	(3)	(0.350)			
16	0					
17	1					

[a] The sum of the products of the S_x and m_x values from age 1 to 15 years.

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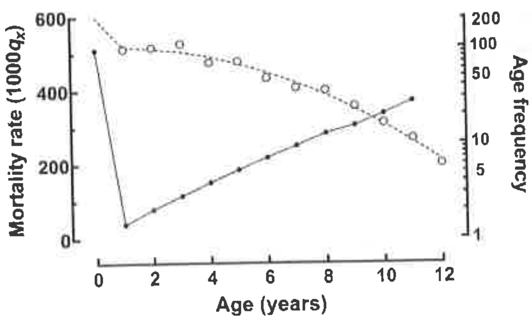

Figure 6.4 Observed age frequencies (open circles) plotted on a logarithmic scale, with a fitted polynomic curve (S_x values in Table 6.3), and age-specific mortality rates (1000q_x) of adjusted values of female tahr. (Corrected data from Caughley 1966. Copyright © 1966. Reprinted by permission of John Wiley & Sons.)

that the observed frequency of females in some age classes is larger than the frequency in younger age classes, i.e. between 1 and 3 years of age, and then again between 4 and 5 years of age. Caughley explained the fluctuations on the basis of sampling error rather than population fluctuations, and fitted a polynomial function to the observed frequencies to ensure that the numbers declined with age. He made a small error in computing his polynomial function and a larger typographical error in his life table (see Simulations website for details) and these have been corrected in Table 6.3 and Figure 6.4. The errors had little influence on his results or conclusions. Mortality is high for newborns, declines to a minimum at about the time of puberty and thereafter steadily increases with age (Figure 6.4). The shapes of the survivorship and mortality rate curves are rather similar to the female large cactus finch (Figure 6.1).

In recent years, this type of life table has become more general (i.e. with much less restrictive assumptions) by combining age-structure data at different times with other demographic data (e.g. population growth rates). This allows the estimation of age-specific survival rates even for populations that are fluctuating in size (Udevitz and Gogan 2012). Note that large unbiased samples of the population are still required.

Stationary Life Tables Based on the Ages at Death Within a Population

This method relies on the ability to retrieve, and reliably age, individuals that have died from natural causes. This is a daunting task for most species because dead individuals may be totally consumed by predators and scavengers, or may be exceedingly difficult to find. When applying this method, there is a strong risk of collecting systematically biased samples, e.g. in large game animals, the survival of the remains of dead newborns is much lower than that of adults. Finally, the method assumes that the population is stationary (i.e. births balance deaths), but we can only determine this by studying the living population.

This method has been used to analyse Adolph Murie's (1944) data on Dall mountain sheep (*Ovis dalli*) in Mount McKinley (now Denali) National Park in Alaska. Murie made two collections of skulls: 608 skulls of sheep that had died before 1937, and 221 skulls of sheep that died between 1937 and 1941. He estimated the age of individuals using their teeth and the growth rings on their horns, and identified the sex of animals that were at least 2 years old from the size of their horns. Life tables were developed using Murie's pre-1937 data by Deevey (1947) and the 1937–41 data by Taber and Dasmann (1957). Unfortunately, the life tables

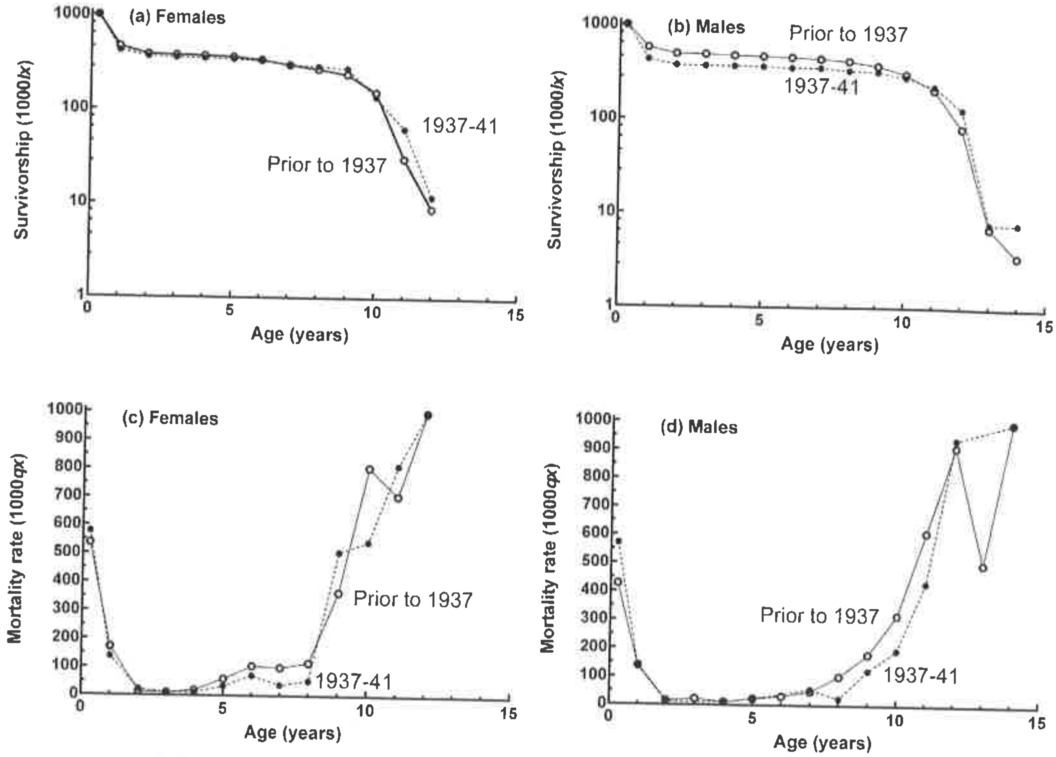

Figure 6.5 (a) Female survivorship, (b) male survivorship, (c) female age-specific mortality and (d) male age-specific mortality for Dall sheep (*Ovis dalli*), estimated from the ages of skulls collected prior to 1937, and between 1937 and 1941. (Data from Murie 1944.)

cannot be compared directly because Deevey combined the two sexes whereas Taber and Dasmann kept the sexes separate and also attempted to correct for the obvious underrepresentation of lamb skulls (aged 0–5 years) in the collections.

I have completely reanalysed the two data sets (pre-1937 and 1937–41) using a modification of Taber and Dasmann's method that is described in detail in the Simulations website for Chapter 6 (Dall sheep). Briefly, for each time period and sex, there were a number of skulls that could not be aged and they were assigned to the older age classes according to their relative frequency. Secondly, the number of newborn sheep was estimated from the age structure of mature females multiplied by 0.8, assuming that all adult females produced an average of 0.8 lambs per year. This is similar to the method used by Caughley (1966).

The survivorship curves and mortality rates of the two data sets are remarkably similar (Figure 6.5). This is surprising because the data sets have significantly different age structures: the proportion of 2–8-year-old adults was higher prior to 1937 compared with 1937–41 (see Simulations website for details), which means that the basic assumption of a stationary age distribution has been violated. Nevertheless, although one would have little confidence in the actual age-specific survival and mortality rates, the general pattern of mortality is probably correct. Survival was slightly better in males than in females: the oldest males were 14 years old whereas the oldest females were 12 years old (Figure 6.5a, b). The mortality rate was high at first, dropped rapidly to its lowest level at 2 years of age, increased slowly to the age of 8 years and thereafter increased rapidly until all of the population was dead (Figure 6.5c, d).

6.3 COMPARISON OF LIFE TABLES

Survivorship curves are often categorized into three types: in type I, the chance of survival decreases with age (e.g. in some human populations, Figure 6.3a); in type II, survival is constant with age (e.g. in Belding's ground squirrel (*Spermophilus beldingi*); Sherman and Morton 1984); and in type III, the chance of survival increases with age (e.g. forest canopy tree species where there is heavy mortality of seedlings). However, survivorship curves rarely conform exactly to these 'ideal' types; they are usually a composite of types (e.g. see Figure 6.1a for *Geospiza conirostris*).

We should also remember that survival is not just a function of age. A study of bighorn sheep (*Ovis canadensis*) in Canada showed that large body mass generally increased the survival of lambs, yearling females, and females of 7 years of age and older, but these effects only became evident at high population density (Festa-Blanchet *et al.* 1997).

Finally, there is good reason to suspect that the shape of age-specific mortality rate curves may be subject to selection (i.e. they have evolved), and we will consider briefly how this may occur in Chapter 8.

6.4 Summary and Conclusions

Life tables document information about age-specific death rates and survival rates in a very organized way. They are typically used for animals that live for a few or many years, but generally not for plants because of their phenotypic plasticity. There are four basic types: (1) cohort life tables, the most accurate, which follow the survival or mortality of a cohort of newborn young from birth until all have died; (2) time-specific or vertical life tables, which collect the information for all age classes at one period of time, which is a practical approach for very long-lived animals; (3) stationary life tables that are based on the age structure of the population; and (4) stationary life tables that are based on the ages of death. This chapter has shown how each of these types of life table can be developed, usually with field observations, and the assumptions and potential errors are discussed.

The patterns of mortality with age can be compared between the sexes, in different environments and between species.

7 Growth of Age-Structured and Stage-Structured Populations

Age-specific birth rates can be estimated by observing the number of live offspring produced by samples of females in different age classes. For example, if a sample of 30 2-year-old females gave birth to 120 live offspring during the course of the year, the age-specific birth rate (B_x) of females in age class 2–3 years would be 120/30, or 4.0 per year. Some animals are secretive, and here we may collect samples, age the individuals, and count the number of eggs, embryos or placental scars per female to estimate birth rates. However, not all the eggs or embryos may be viable, which will result in the fertility being overestimated. The fertility of many animals, and particularly plants, is more related to their size rather than age and so their population growth rates are better estimated from the size structure of the population. Clearly, there are many practical issues that need to be considered when estimating population growth rates.

7.1 POPULATION GROWTH RATES IN AGE-STRUCTURED POPULATIONS

Population growth rates are calculated by combining the information on age-specific birth rates with the age-specific death rates derived from cohort life tables (see Chapter 6). Note that stationary life tables should not be used to calculate population growth rates for obvious reasons (r is assumed to equal zero).

There are three ways of estimating population growth rates from the age structure of populations, which we will illustrate by way of a classical study on the Uinta ground squirrel (*Spermophilus armatus*) in Utah, USA (Slade and Balph 1974). The ground squirrels were studied intensively for 7 years on the grounds of a field station where there was a central area of lawn (0.9 ha) surrounded by 8.0 ha of mixed trees, shrubs and grasses (non-lawn). All squirrels were captured and marked as they emerged from hibernation from late March to mid-April and their young were also captured and marked as they emerged from their natal burrows from late May to early June. The entire population was monitored on a daily basis until most of the population disappeared underground in late summer. From 1964 until early 1968, the population was left undisturbed and data were collected on age-specific birth and death rates. Then, in 1968, the population was reduced by approximately 60 per cent in a manner that did not change the age and sex composition of the population. In 1969 and 1970, the population density was kept low by removing 40 per cent of the emerging juveniles and the resulting

Table 7.1 **Age-specific survival (l_x) and live female births per female of age x (m_x) for Uinta ground squirrels (*Spermophilus armatus*) in lawn and non-lawn habitats in Utah, before and after the approximate halving of population density. The population growth parameters are also calculated (see text and Table 7.2)**

	Lawn				Non-lawn			
Age (years)	Pre-reduction		Post-reduction		Pre-reduction		Post-reduction	
x	l_x	m_x	l_x	m_x	l_x	m_x	l_x	m_x
0	1.000	0.00	1.000	0.00	1.000	0.00	1.000	0.00
0.75	0.292	1.96	0.359	1.75	0.375	0.94	0.474	1.67
1.75	0.128	2.73	0.190	2.76	0.157	1.77	0.228	2.04
2.75	0.041	2.73	0.089	2.76	0.079	1.77	0.134	2.04
3.75	0.013	2.73	0.041	2.76	0.039	1.77	0.079	2.04
4.75	0.000	–	0.019	2.76	0.020	1.77	0.046	2.04
5.75			0.009	2.76	0.010	1.77	0.027	2.04
6.75			0.000	–	0.000	–	0.016	2.04
R_0	1.069		1.589		0.892		1.873	
T_c	1.386		1.813		1.865		2.001	
r_c	0.048		0.255		−0.061		0.314	
r_m	0.049		0.282		−0.060		0.375	

From Slade and Balph (1974). Reprinted with permission of John Wiley & Sons.

changes in age-specific birth and death rates were observed. Overall, the population density in the last 2 years (post-reduction) was approximately half that of the pre-reduction population density. The basic data and calculated population growth rate parameters before and after population reduction in the two habitats are presented in Tables 7.1 and 7.2. I have omitted the data for squirrels living on the perimeter of the lawn that utilized both habitats (Slade and Balph 1974).

The production of new females for each age class is calculated by multiplying the survivorship from birth (l_x) by the average number of live female births (m_x) per female of age x (Table 7.2). The sum of the $l_x m_x$ series estimates the net reproductive rate, R_0, for the population, which is the average production of female offspring per female during her lifetime. Thus:

Table 7.2 **Calculation of population growth parameters for Uinta ground squirrels (*Spermophilus armatus*) inhabiting the lawn habitat prior to the reduction in population density (see text and Table 7.1)**

Age (x) (years)	l_x	m_x	$l_x m_x$	$x l_x m_x$	Trial r_m value	$e^{-r_m x} l_x$	$e^{-r_m x} l_x m_x$
(1)	(2)	(3)	(4) = (2) × (3)	(5) = (1) × (2) × (3)	(6)	(7) = ($e^{-(6)\times(1)}$) × (2)	(8) = (7) × (3)
0	1.000	0.00	0.000	0.000	0.049	1.0000	0.0000
0.75	0.292	1.96	0.572	0.429	0.049	0.2815	0.5517
1.75	0.128	2.73	0.349	0.612	0.049	0.1175	0.3207
2.75	0.041	2.73	0.112	0.308	0.049	0.0358	0.0978
3.75	0.013	2.73	0.035	0.133	0.049	0.0108	0.0295
4.75	0.000	–	–	–			
Sum			1.069	1.482			0.9997

$R_0 = \Sigma l_x m_x = 1.069$

$T_c = \Sigma x l_x m_x / \Sigma l_x m_x = 1.482/1.069 = 1.386$

$r_c = (\ln(R_0))/T_c = 0.048$

$r_m = 0.049$ (when $\sum e^{-r_m x} l_x m_x = 1$)

Note that the information in Table 7.1 is for the female segment of the population. This is typical for dioecious species where we assume that the male segment will grow at the same rate. In monoecious species, we would provide data for the entire population.

$$R_0 = \Sigma l_x m_x \qquad \text{(Eqn 7.1)}$$

Prior to the reduction of population density, the average lawn female entering the population produced 1.069 female offspring before she died, whereas after the population density was halved, a female replaced herself with 1.589 female offspring before she died, a substantial increase (Table 7.1). Note that in the non-lawn areas, the replacement rate more than doubled after the population density was reduced.

The net reproductive rate (R_0) is a measure of the multiplication rate (λ) per generation (see Chapter 4). If $R_0 = 1.0$, the population is just replacing itself and is neither increasing nor decreasing, i.e. it is stationary; if $R_0 > 1$ the population is increasing in size; and if $R_0 < 1$ the population is decreasing in size. In order to calculate a growth rate, r, per unit time we need to be able to estimate the generation time and there are two basic ways of doing this.

7.1.1 The Capacity for Increase (r_c)

In Chapter 4, we showed that for populations with constant growth rates, r, the population size at time t (N_t) is predicted by Eqn 4.7, i.e. $N_t = N_0 e^{rt}$, and so after one generation, where $t = T$ (the generation time), the equation becomes $N_T = N_0 e^{rT}$. If we rearrange this equation by dividing both sides by N_0 and taking the natural logarithm of the expression, we obtain $\ln(N_T/N_0) = rT$. But $N_T/N_0 = R_0$ and so the last equation can be rearranged to:

$$r_c = \frac{\ln(R_0)}{T} \quad \text{(Eqn 7.2)}$$

where r_c is the capacity for increase, and $R_0 = \Sigma l_x m_x$.

To estimate r_c, we estimate the generation time (T) by calculating the average age at which the females produce their offspring (i.e. the mean age of the $l_x m_x$ series). This is calculated by weighting the production of offspring by the age at which they are produced (i.e. the sum of the $x l_x m_x$ series), and dividing that value by the number of offspring produced (i.e. the sum of the $l_x m_x$ series). Symbolically, this is:

$$T = \frac{\sum x l_x m_x}{\sum l_x m_x} \quad \text{(Eqn 7.3)}$$

This estimates an average parental age at which all the offspring are born, and is equivalent to what would happen if all of the offspring were born to females at that age. You can check that this equation gives the correct estimates of the generation time for our example in Table 7.2. For a more complete set of calculations for the population growth parameters in Table 7.1, go to Chapter 7 of the Simulations website.

The r_c values for the population living in the two habitats, before and after the reduction in density, are given in Table 7.1. The multiplication rates per year (λ) are calculated by taking the exponents of these estimates. Prior to the population being reduced, the multiplication rate was 1.049 for the lawn area and 0.941 for the non-lawn area. Thus, the population living on the lawn was growing at about 5 per cent per year and that living in the non-lawn area was declining at about 6 per cent per year. Slade and Balph (1974) suspected that the excess individuals from the lawn area migrated to the non-lawn area, keeping the overall population more or less in balance. After the population density was approximately halved, the growth rate (λ) dramatically increased to 1.29 in the lawn area and 1.369 in the non-lawn area. We will analyse the reasons for this in Section 7.2.1.

7.1.2 The Intrinsic Rate of Natural Increase (r_m) and the Euler Equation

The second, and more accurate, method for estimating exponential growth rates of age-structured populations makes use of the Euler equation, which was first developed in the eighteenth century by a Swiss mathematician, Euler (pronounced 'Oiler'), in his analysis of human demography. It was independently derived by Alfred Lotka, at the beginning of the twentieth century, who showed the equation's utility for demographers. The basic equation is as follows:

$$1 = \int_0^\infty e^{-r_m x} l_x m_x . \partial x \qquad \text{(Eqn 7.4)}$$

Its equivalent discrete form is given by:

$$1 = \sum_0^\infty e^{-r_m x} l_x m_x \qquad \text{(Eqn 7.5)}$$

When an age-structured population increases exponentially, each age class is increasing at the same rate (r_m) and the age structure of the population stabilizes into a *stable age distribution* that is unique to the particular combination of survivorship (l_x) and fertility (m_x) values of the population. The Euler equation uses this fact to calculate the true rate of exponential growth (r_m) for the population. This is illustrated in Table 7.3, which shows why Eqn 7.5 is structured the way it is.

The first column in Table 7.3 represents a time prior to time 0 (t_0) for the second column, but for the remainder of the table, the column represents age (x) in the life table. In column 2, we calculate the number of individuals that would have been born at previous time steps

Table 7.3 **A numerical example to illustrate the development of the Euler equation (Eqn 7.5). The trial value of r is 0.1, and the survivorship (l_x) and fecundity (m_x) series are as shown. See text for further explanation**

(1)	(2)	(3)	(4)	(5)	(6)
Time prior to time 0 or age (x)	Number of individuals born at time x (e^{-rx})	(l_x)	No. individuals surviving to time 0 ($e^{-rx}l_x$) = (2) × (3)	(m_x)	Number of offspring born to surviving individuals at time 0 ($e^{-rx}l_x m_x$) = (4) × (5)
0	1.0	1.0	$l_0 = 1.0$	0	0
1	$e^{-0.1} = 0.9048$	0.8	(2) × 0.8 = 0.7238	0.5	(4) × 0.5 = 0.3619
2	$e^{-0.2} = 0.8187$	0.6	(2) × 0.6 = 0.4912	1.0	(4) × 1.0 = 0.4912
3	$e^{-0.3} = 0.7408$	0.4	(2) × 0.4 = 0.2963	0.4	(4) × 0.4 = 0.1185
4	$e^{-0.4} = 0.6703$	0.2	(2) × 0.2 = 0.1341	0.21	(4) × 0.21 = 0.0282
5	–	0	–	0	–
					$\Sigma = 0.9998$

if the population growth rate (r) remained constant. In our example, we have selected a trial r value of 0.1. For the previous times 1, 2, . . ., x, this is equal to $e^{-0.1}$, $e^{-0.2}$, . . ., e^{-rx}, which in our example develops a series of female births in succeeding time steps that increase exponentially from 0.6703 to 1.0, at a growth rate of 0.1. Then, in column 4, we calculate the number of survivors at time 0 of the calculated births in column 2 multiplied by their probability of survival from birth in column 3. Of the 0.9048 individuals born in the previous time step, only 80 per cent survive to the present time (0), and so the proportion in age class 1 is 0.9048 × 0.8 = 0.7238. Thus, the series is calculated using the formula $e^{-rx}l_x$ (in column 4). Now that we know the number of individuals in each age class (column 4), we only need to multiply this by the age-specific fertility rate (m_x) in column 5 to obtain the total number of female offspring at time 0 ($\Sigma e^{-rx}l_x m_x$) in column 6. In our example, the total is 0.9998, which is sufficiently close to 1.0 to conclude that our trial r value of 0.1 is approximately equal to r_m according to the Euler equation (Eqn 7.5). Column 4 represents the stable age distribution in the form of a modified survivorship series. Normally, the stable age distribution is given as the proportion in each age class of the population, which is calculated by the following equation:

$$\frac{e^{-rx}l_x}{\sum e^{-rx}l_x} \qquad \text{(Eqn 7.6)}$$

In our example in Table 7.3, the stable age distribution for individuals aged 0–4 is 0.3780, 0.2736, 0.1857, 0.01120 and 0.0507, respectively.

Of course, we could have gone through our series of calculations only to find that the sum of $e^{-rx}l_x m_x$ does not equal 1 and so our trial r value does not equal r_m and therefore we have not calculated the stable age distribution. In this case, we would have to adjust the trial r value and repeat the set of calculations until we hit the correct one. This is tedious if one only has a calculator but is much simpler using a spreadsheet. Go to Chapter 7 of the Simulations website to see the procedure for calculating r_m and the stable age distribution for the Uinta ground squirrel data in Table 7.1.

7.1.3 Comparison of r_c and r_m

Most authorities consider that r_c is simply an approximation for r_m, but there is a precise relationship between the two estimates that has been elucidated by Laughlin (1965). If there is no overlapping of generations, such as is seen in annual plants and many insects, then $r_c = r_m$. However, where there is overlapping of generations and the population has a stable age distribution (see Section 7.1.2), r_m calculates the true rate of population growth and r_c is an underestimate. This is because, for a population with a stable age distribution, the generation time (T) is overestimated by Eqn 7.3. The difference between r_m and r_c becomes greater as the growth rate (r) increases. We can see this in our example of the Uinta ground squirrel (Table 7.1). The pre-reduction values of r_c are only about 2 per cent lower than the r_m values in the two habitats, whereas for the post-reduction populations the r_c values are between

10 per cent and 16 per cent lower than the r_m values. It should be emphasized that populations will only grow at a rate of r_m when they have attained a stable age distribution (the age distribution when all age classes are increasing in number at the same rate). If a population with overlapping generations is introduced into a new area and increases in number, or if a population has been stationary for a time and then starts to increase in number, the rate of increase during the initial phase will be closer to r_c than to r_m, which is likely in our example of Uinta ground squirrels. This is because it takes time for the population to attain a stable age distribution.

7.1.4 A Matrix Model of Age-Structured Population Growth

The growth of age-structured populations can also be modelled using matrix algebra. Caswell (2001) provides an excellent overview of this area. My purpose here is not to worry about matrix algebra *per se* but to show why these models are important. I will use some of the data from Table 7.1 to show how the model is developed, but in order to do this we need to make one change. The Uinta ground squirrel breeds at the same time every year and so it is not possible that breeding first begins after 0.75 years and thereafter at yearly intervals. Consequently, the age sequence of 0, 0.75, 1.75, 2.75, etc. will be changed to 0, 1, 2, 3, etc. This is also necessary because in the matrix model the age intervals must have a constant length.

The model consists of a *transition* or *population projection matrix* [A], which defines how the population moves from one age class to another (see Figure 7.1 and Box 7.1). Matrix [A] is often referred to as a *Leslie matrix* (after one of the originators of the method), and is a $k \times k$ square matrix, where k is equal to the number of age classes in the population. The model, as applied to the Uinta ground squirrel, is outlined in Box 7.1. Matrix [A] has a characteristic structure where the first row shows the per capita production of offspring in each age class, and this is multiplied by the number in each age class at time t, i.e. $F_1 \times n_1, F_2 \times n_2, \ldots, F_5 \times n_5$, and the sum of these calculations equals n_1 at time $t + 1$. The probabilities of surviving from one age class to the next (P_i) are shown as a diagonal sequence in columns 1–4 in rows 2–5 in matrix [A]. P_1 in column 1 is the probability of surviving from n_1 to n_2 and is multiplied by n_1

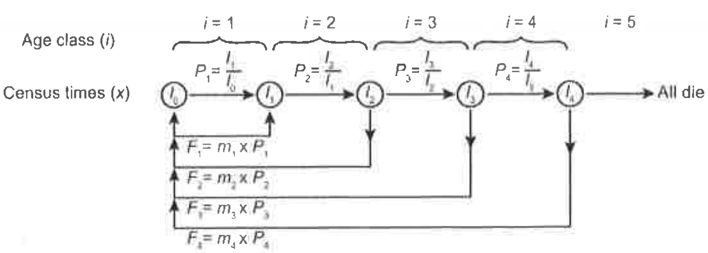

Figure 7.1 Schematic diagram of a matrix model for a population with five age classes (i). There are five census times (x) starting at birth (l_0) to 4 years of age (l_4), which corresponds to a cohort survival series. Breeding is seasonal and occurs just prior to the census times, corresponding to a birth-pulse, post-breeding census matrix model (Caswell 2001). See Table 7.4, Box 7.1 and text.

Table 7.4 **Age-specific survival probabilities (P_i) and fertilities (F_i) of the Uinta ground squirrel for the pre-reduction population living in the lawn habitat. Data from Table 7.1 with modified census ages (see text)**

Age (x)	l_x	m_x	Age class (i)	$P_i = \frac{l_{x+1}}{l_x}$	$F_i = m_x P_i$
0	1.000	0			
			1	0.292/1.000 = 0.292	1.96 × 0.292 = 0.572
1	0.292	1.96			
			2	0.128/0.292 = 0.438	2.73 × 0.438 = 1.197
2	0.128	2.73			
			3	0.041/0.128 = 0.320	2.73 × 0.320 = 0.874
3	0.041	2.73			
			4	0.013/0.041 = 0.317	2.73 × 0.317 = 0.866
4	0.013	2.73			
			5	0/0.013 = 0	0 × 0 = 0
5	0	0			

Data from Table 7.1 with modified census ages (see text).

at time t to calculate the number of n_2 at time $t + 1$, P_2 in column 2 is the probability of surviving from n_2 to n_3 and is multiplied by n_2 at time t to calculate the number of n_3 at time $t + 1$, and so on to P_4 in column 4, which is the probability of surviving from age class 4 to 5. Note that there is no P_5 because none of the population survives to the sixth age class.

We can use this matrix model to show how a population grows and how its age structure changes over time. To see how this can be done using Excel, go to the Simulations website for Chapter 7 on the Uinta ground squirrel. If we start with the same number of individuals in each age class, we can see that, after some initial fluctuations that were quickly damped, all age classes grew at the same exponential rate (Figure 7.2). At this point, a stable age distribution has developed and the growth rate, r, has stabilized at the value of r_m. Growth was positive for the lawn population and negative for the non-lawn population, as indicated in Table 7.1.

We should note that the values of r_m and the stable age distribution calculated by the Euler equation (Section 7.1.2) are identical to those calculated using the Leslie matrix (see Simulations website). However, those calculated in Section 7.1.2 are not the same as those in our matrix model because we modified the age sequence from 0, 0.75, 1.75, 2.75, etc. to 0, 1, 2, 3, etc., as explained at the start of this section.

BOX 7.1 AGE-STRUCTURED MATRIX MODEL OF POPULATION GROWTH

Lawn Population of Uinta Ground Squirrel (see Figure 7.1)

The model takes the form:

$$n(t+1) = [\mathrm{A}] \times n(t) \qquad \text{(Eqn 7.7)}$$

where the number in each age class is calculated from the number in the previous age class pre-multiplied by matrix [A], which defines how the population moves from one age class to another. The parameters in matrix [A] are estimated differently according to the situation (Caswell 2001). The Uinta ground squirrel breeds once a year and was censused just after the birth of the young, corresponding to Caswell's *birth-pulse, post-breeding census* formulation. The lawn population has five age classes, and so Eqn 7.7 takes the form:

$$\begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix}(t+1) = \begin{bmatrix} F_1 & F_2 & F_3 & F_4 & F_5 \\ P_1 & 0 & 0 & 0 & 0 \\ 0 & P_2 & 0 & 0 & 0 \\ 0 & 0 & P_3 & 0 & 0 \\ 0 & 0 & 0 & P_4 & 0 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix}(t)$$

From Table 7.4, $F_1 = 0.572$, $F_2 = 1.197$, $F_3 = 0.874$, $F_4 = 0.866$ and $F_5 = 0$, and $P_1 = 0.292$, $P_2 = 0.438$, $P_3 = 0.320$ and $P_4 = 0.317$ (note there is no P_5 because it equals 0).

The probability of survival from one age class to the next (P_i) is simply the l_x value at the start of the age class divided into the l_x value at the start of the following age class:

$$P_i = \frac{l_{x+1}}{l_x} \qquad \text{(Eqn 7.8)}$$

Females give birth at the end of each age class because of the post-breeding census, and so fertility coefficients (F_i) are calculated by multiplying the m_x values by the probability of survival (P_i) during the age class prior to breeding:

$$F_i = m_x P_i \qquad \text{(Eqn 7.9)}$$

The matrix calculations proceed as follows: in matrix [A], the values in the first column multiply n_1 of matrix $n(t)$, those in the second column multiply n_2 of matrix $n(t)$, and so on to those in the fourth column that multiply n_4 in matrix $n(t)$. (Note that the values in the fifth column of matrix [A] are all zeros because all n_5 individuals in matrix $n(t)$ die.) The results of the matrix [A] $\times$ matrix $n(t)$ calculations are summed by row (row 1–5) and transferred to the corresponding row in matrix $n(t + 1)$.

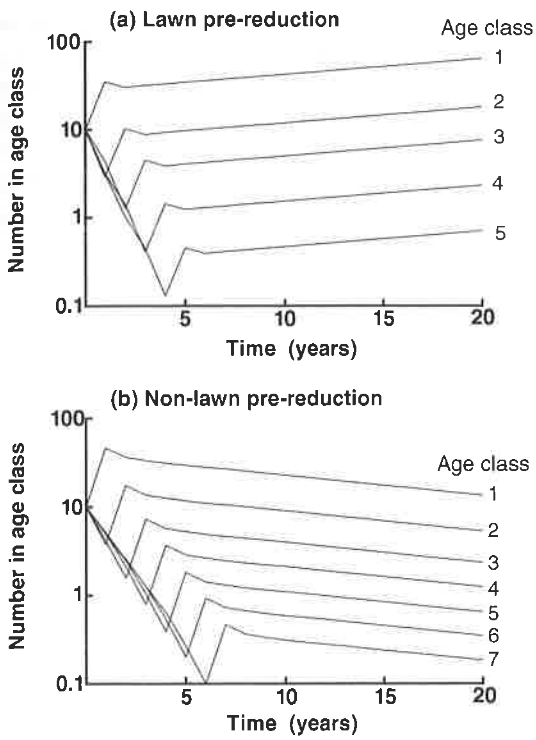

Figure 7.2 Matrix projection of the growth of the different age classes for the pre-reduction populations in (a) lawn areas and (b) non-lawn areas, to show that they rapidly develop a stable age distribution. Data from Table 7.1 as modified for the matrix model (see text).

7.2 ANALYSING THE INFLUENCE OF DEMOGRAPHIC VARIABLES ON POPULATION GROWTH RATES

The Leslie matrix can also be analysed using matrix algebra to assess the relative effects of demographic variables (e.g. age at first reproduction, survival and fertility at different ages) on population growth rates (Caswell 2001). For example, each demographic variable could be changed by a given amount (e.g. 10 per cent) to see which ones cause the largest change in growth rate (λ). Such prospective changes can be compared with the results of *life table response experiments* where natural or experimentally induced environmental changes (e.g. habitat, food supplementation, population density) cause actual changes in demographic variables.

7.2.1 Effect of Density Reduction on Uinta Ground Squirrels

Slade and Balph (1974) experimentally reduced the ground squirrel populations in the lawn and non-lawn habitats by about 50 per cent (see Section 7.1.1), and the populations responded by dramatically increasing their multiplication rates (λ). This represents a life table response experiment on the population, which has been analysed by Oli *et al.* (2001). They modified the life tables (Table 7.1) in the same way as in Section 7.1.4; i.e. by changing the age (x) series from 0, 0.75, 1.75, etc. to 0, 1, 2, etc., and then calculated the survival probability (P_i) and

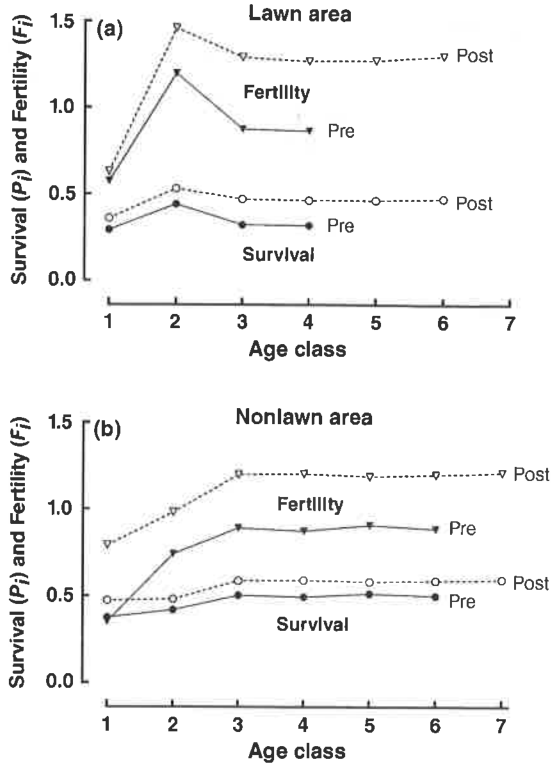

Figure 7.3 Graphs showing that the reduction of population density increased both survival and fertility of the Uinta ground squirrel. Pre, pre-reduction; post, post-reduction. Data from Table 7.1 with the age distribution modified to conform to the matrix model, and the survival probabilities and fertilities calculated as illustrated in Table 7.4.

fertility (F_i) for the young of the year (age class 1), yearlings (age class 2) and adults (age class 3 and older). It was only necessary to calculate single P_i and F_i values for adults because they were virtually constant with age (see Figure 7.3). They then analysed the potential and actual effects of the following demographic variables on the population growth rate: age at maturity, age of last reproduction, juvenile survival (the first two age classes), adult survival and fertility rate.

The matrix model showed that the increase in λ in the lawn habitat was *potentially* most affected by reducing the age of first reproduction and secondly by increasing the fertility rate, but the order of these two variables was reversed in the non-lawn habitat. However, the *actual* increase in λ was mainly a result of increasing the fertility rate followed by increasing juvenile survival in both habitats (Figure 7.3). The age of first reproduction did not change in response to the density reduction. Yearling females reproduce soon after they emerge from an 8-month winter hibernation period. They could only reduce the age of maturity if they bred in the summer months prior to hibernation, but they need to grow and build sufficient fat reserves for hibernation during this period. Consequently, the species is constrained in reducing its age of maturity, a feature that is probably common in seasonal breeders that live in highly seasonal environments.

The matrix analysis of demographic variables by Oli *et al.* (2001) failed to show that the *actual* increase in fertility rate occurs differently in the two habitats. This is because fertility is

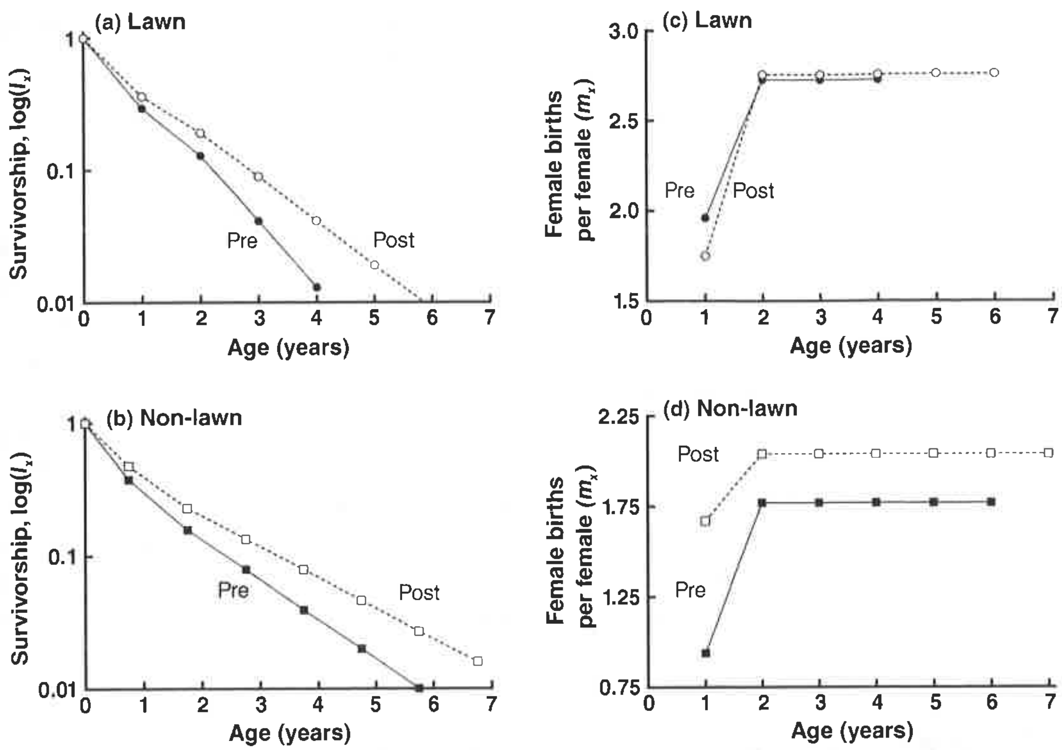

Figure 7.4 Examination of the causal factors for the increase in fertility ($F_i = P_i m_x$) when population density is reduced (see Figure 7.3 and text). Pre, pre-reduction; post, post-reduction.

the product of two variables ($F_i = m_x P_i$; Eqn 7.9). In the lawn area, m_x shows no general increase after population reduction (Figure 7.4c) and so the increase in fertility is purely a result of increased survival, or P_i (Figure 7.4a). In contrast, in the non-lawn area, the fertility increased because both m_x and P_i increased (Figure 7.4b, d). What habitat-related variables might explain this difference in response? Females preferred the lawn habitat, occurring there in the highest density, and the mowed grass provided an abundance of food during the mating season, which resulted in a high birth rate. Presumably, when the density of the population was halved, the additional food that became available could not be utilized effectively to increase the birth rate and so there was a trivial change in m_x. The presence of trees and bushes reduced the abundance of grasses and herbs in the non-lawn habitat, and as a consequence the m_x values were much lower than in the lawn habitat. However, the increased availability of food following density reduction did lead to a substantial increase in birth rates. Towards the end of the growing season, the non-lawn habitat became advantageous because of the presence of seeds, which were lacking in the lawn habitat. Seeds were a preferred food because they are particularly important in building up fat reserves prior to hibernation. Perhaps this is why survival was consistently higher in the non-lawn habitat compared with the lawn habitat (Table 7.1, Figure 7.4a, b). In conclusion, an analysis of the importance of various demographic variables using matrix algebra techniques must be done with care. The variables must be chosen carefully; otherwise they can give misleading results.

7.3 ANALYSING STAGE-CLASSIFIED LIFE CYCLES

We have used the age-specific birth and death rates to calculate population growth rates of a population but this may not be appropriate for some organisms. For example, many insects go through a series of developmental stages (egg, larva, pupa, and adult) that may vary in duration according to food availability, temperature, and moisture. In some plants and fish, the capacity for sexual reproduction may be more influenced by size and developmental stage than by chronological age. Thus, an immature, intermediate-sized plant or fish may be a fast-growing juvenile or a stunted 'adult'. To complicate matters further, many plants will regress to a juvenile stage if they are subject to herbivory. In cases such as these, it may be difficult or impossible to calculate the growth rate of the population using the more conventional Euler equation, but it is relatively simple to develop a stage- or size-classified matrix model that can be analysed (Caswell 2001). An explanation of stage-structured matrix models based on the following *Trillium* example is provided in Box 7.2.

7.3.1 Population Dynamics of *Trillium grandiflorum* (Liliaceae)

The herbaceous perennial *Trillium grandiflorum* inhabits the understory of deciduous forests in eastern North America. Tiffany Knight (2004) has studied the demography in 12 populations of this lily in north-western Pennsylvania. The individuals are non-clonal and come into leaf in late April and senesce in late July, and then persist in a dormant state underground until the following year. Knight identified six demographic changes: germinant (Germ), which are germinating seeds with a root but no above-ground shoot, seedlings (SL), one-leaf (1L), small three-leaf (S3L), large three-leaf (L3L) and reproductive (Rep), which has a single flower and seed capsule. By following the fates of hundreds of marked individuals from year to year, and by performing various experiments, she has estimated the transition probabilities between the six demographic stages from one year to the next for each population (Figure 7.5). *Trillium* is a preferred food of white-tailed deer (*Odocoileus virginianus*), and Knight has studied their impact on this woodland lily.

Effects of Herbivory and Pollen Limitation

Herbivory and pollen limitation both reduce the population growth rate, and Knight studied their relative effects on a single population where both factors were consistently high (Knight 2004). Only the last two stages, large three-leaf (L3L) and reproductive (Rep), were eaten by deer, which clipped the plant just below the whorl of three leaves, removing them and any flower or fruit present. Note that the transition probabilities (Figure 7.5) where there was herbivory (H) included all plants whether they were eaten or not, and so measured the effects

BOX. 7.2 STAGE-STRUCTURED MATRIX MODEL OF POPULATION GROWTH

Population Growth of *Trillium* at Ambient Levels of Herbivory and Pollen Limitation

The growth of this species is described in Section 7.3.1, and is based on the data of Knight (2004). The values in the transition matrix below are taken from Figure 7.5:

$$\begin{array}{c} n(t+1)\text{ values} \\ \begin{bmatrix} \text{Germ} \\ \text{SL} \\ \text{1L} \\ \text{S3L} \\ \text{L3L} \\ \text{Rep} \end{bmatrix} \end{array} (t+1) = \begin{array}{c} \begin{array}{cccccc} \text{Germ} & \text{SL} & \text{1L} & \text{S3L} & \text{L3L} & \text{Rep} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 3.55 \\ 0.51 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.31 & 0.5 & 0.04 & 0 & 0 \\ 0 & 0 & 0.32 & 0.82 & 0.33 & 0 \\ 0 & 0 & 0 & 0.09 & 0.58 & 0.61 \\ 0 & 0 & 0 & 0 & 0.09 & 0.39 \end{bmatrix} \end{array} \begin{array}{c} n(t)\text{values} \\ \begin{bmatrix} \text{Germ} \\ \text{SL} \\ \text{1L} \\ \text{S3L} \\ \text{L3L} \\ \text{Rep} \end{bmatrix} \end{array} (t)$$

The transition matrix [A] is more complex than that of the age-structured matrix model in Box 7.1 because at some stages the source of individuals may be by development from a previous stage, by regression from a later stage or by individuals remaining at the same stage for more than 1 year. Nevertheless, the same rules of matrix algebra apply as in age-structured matrices (Box 7.1).

Germinants at time $t + 1$ are produced at a rate of 3.55 by the number of reproductive individuals at time t (4.49 seed production $\times$ 0.79 germination rate). Just over half (0.51) of the germinants at time t develop into seedlings at time $t + 1$. The number of one-leaf individuals at time $t + 1$ is the sum of 0.31 $\times$ number of seedlings + 0.5 of one-leaf individuals + 0.04 of small three-leaf individuals at time t. Similarly, the number of small three-leaf individuals at time $t + 1$ is the sum of 0.32 of one-leaf individuals + 0.82 of small leaf individuals + 0.33 of large three-leaf individuals at time t. Likewise, the number of large three-leaf individuals at time $t + 1$ is the sum of 0.09 of small three-leaf individuals + 0.58 of large three-leaf individuals + 0.61 of reproductive individuals at time t. Finally, the number of reproductive individuals at time $t + 1$ is the sum of 0.09 of large three-leaf individuals + 0.39 reproductive individuals at time t. The various matrix analyses for *Trillium* may be accessed at the Simulations website for Chapter 7.

of ambient herbivory on the population. The no-herbivory (NH) transition probabilities simply excluded all plants that were eaten from the data matrix. Plants in a pollen supplementation treatment (no pollen limitation (NPL)) produced on average five more seeds than control plants (i.e. ambient pollen limitation (PL)). The seed production on individual plants was adjusted accordingly in the database in the NPL treatment, and consequently there were four measures of fecundity related to the presence and absence of herbivory and the presence and absence of

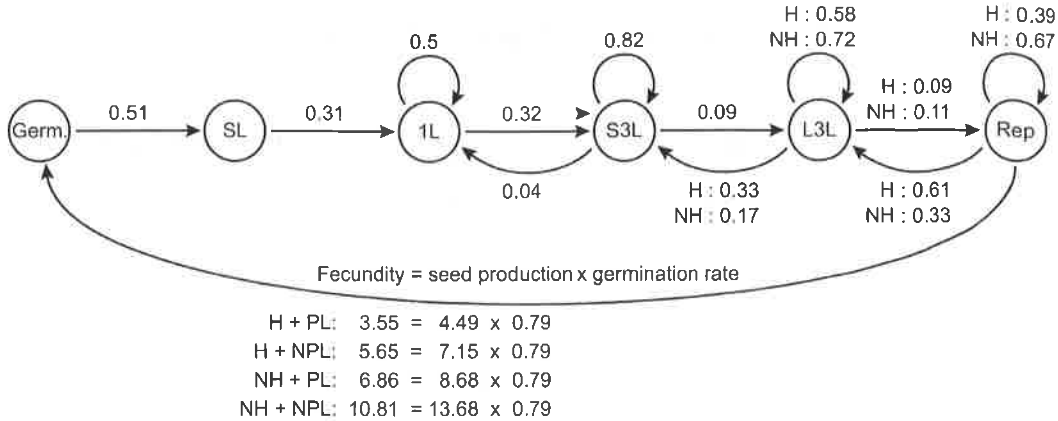

Figure 7.5 Annual probability of transfer between different life stages of *Trillium grandiflorum* under four conditions: ambient herbivory and pollen limitation (H + PL), ambient herbivory and no pollen limitation (H + NPL), no herbivory and pollen limitation (NH + PL), and no herbivory and no pollen limitation (NH + NPL). Life stages: underground germinant (Germ), seedling (SL), one-leaf (1L), small three-leaf (S3L), large three-leaf (L3L) and reproductive (Rep).
(A condensed version from Knight 2004. Copyright © 2004. Reprinted by permission of John Wiley & Sons.)

pollen limitation. A composite of the four matrices is shown in Figure 7.5, and individual matrix projections are calculated in the Simulations website for Chapter 7 on *Trillium*.

Under ambient conditions of herbivory and pollen limitation (H + PL), this population was projected to decline by 3 per cent per year (λ = 0.968 per year). Removing pollen limitation (H + NPL) had a negligible effect on the population growth rate (λ = 0.975 per year), but eliminating herbivory allowed the population to increase (λ = 1.018 per year with pollen limitation, i.e. NH + PL, and λ = 1.035 per year with no pollen limitation, i.e. NH + NPL; Figure 7.6a). Not surprisingly, the removal of herbivory also resulted in a dramatic shift of the stable age (i.e. stage) distribution (Figure 7.6b, c). The regression of the reproductive stage to large three-leaf plants and large three-leaf plants to small three-leaf plants was almost halved (see Figure 7.5), which dramatically reduced the proportion of small three-leaf plants. The removal of herbivory and pollen limitation both dramatically increased the proportion of new recruits (germinants).

Knight (2004) concluded that herbivory by deer needs to be reduced if this population is to survive over the long term. She also noted that there was a negative correlation between population growth rate and the proportion of plants eaten by deer across the 12 populations she studied, which suggests that the level of herbivory is the major factor influencing the persistence of these populations.

Timing of Herbivory and Evolutionary Dynamics

The impact of herbivory also depends on when it occurs during the growing season. Mature *Trillium* eaten early in the growing season were more likely to regress to a non-reproductive stage than those consumed late in the growing season (Knight 2003).

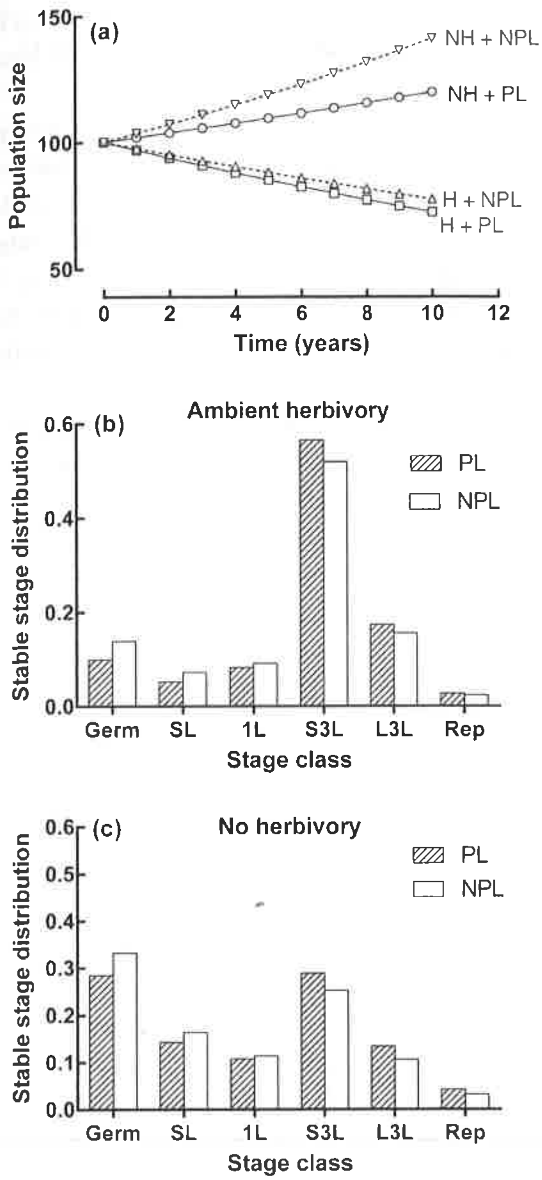

Figure 7.6 (a) Population growth of *Trillium* under four conditions (see Figure 7.5 for abbreviations). (b, c) Stable age distributions of different life stages with and without pollen limitation: under ambient herbivory (b) and without herbivory (c).

(Parts (b) and (c) from Knight 2004. Copyright © 2004. Reprinted by permission of John Wiley & Sons.)

Knight quantified the effects of early and late herbivory on reproductive plants in four populations using a clipping experiment to simulate three levels of herbivory by white-tailed deer (30 plants per treatment per population): clipped early (1 May), clipped late (1 June) or not clipped as a control (Knight 2007). The transition coefficients of reproductive plants from one year to the next for the three treatments were then measured in the normal way. Obviously, clipped plants did not produce seeds in the year they were clipped, but plants clipped early in the season were 2.5 times more likely to regress to a pre-reproductive stage in the following year than late-clipped and control plants. The matrices were analysed to determine the effects of early and late herbivory, and showed that the timing of herbivory was as important as the level of herbivory. The decline in deer herbivory during the flowering season is probably because the later flowers become hidden under the foliage of later emerging species.

Thus, early herbivory by deer reduces the fitness of *Trillium* more than late herbivory, and so assuming that flowering time is heritable, there will be selection to flower later. This evolutionary dynamic was explored by Knight *et al.* (2008) using stage-structured matrix models. The analyses are complex but revealed that selection in some populations could change their demographics from those that are declining very slowly to those that are increasing. Such an evolutionary response would take hundreds of years and is strongly influenced by the initial population size, the initial rate of population decline and the heritability coefficient of flowering time.

In conclusion, the use of matrix models has proved to be a very powerful tool in analysing the population dynamics of many organisms because they are so flexible in their use.

7.4 Summary and Conclusions

This chapter has concentrated on the mechanics of calculating population growth rates where the birth and death rates vary with age, or vary with the developmental stage (e.g. egg, larva, pupa, adult) or size.

For age-structured populations, there are three ways of calculating growth rates. The first is a simple modification of the exponential growth equation (see Chapter 4), which calculates the capacity for increase. This provides a reasonably accurate measure of the growth rates of slowly growing populations. The second method is more accurate for populations with a stable age distribution, and is based on the Euler equation. The third method is a matrix model based on the Leslie matrix, and is extremely flexible and can be used to calculate the growth of any population. For example, it can model the growth of a population as it develops from a stationary age distribution where $r = 0$, to a stable age distribution where growth is exponential. It can also be used to assess the relative effects of demographic variables, such as age of first reproduction or changes in mortality rates, on population growth rates. All of the above are illustrated using data from the classic study of the Uinta ground squirrel by Slade and Balph (1974).

The matrix model is the only method of analysing stage-classified life cycles and can deal with complications such as plants regressing to a juvenile stage if the adult is subject to herbivory. The flexibility of the model was illustrated by analysing the population dynamics of *Trillium grandiflorum* studied by Tiffany Knight.

7.5 Problem

1. The following pattern of mortality was observed in a fictitious mammal. Of the newborn females, half survived for 1 year, 10 per cent survived for 2 years, 3.75 per cent survived for 3 years, 1.5 per cent survived for 4 years and none survived for 5 years. Breeding occurred once a year, and immediately following the breeding season the age-specific reproductive rates were determined from the fresh placental scars of females of known age. One-year-old females had an average of 2.8 placental scars per female, 2-year-old females had an average of 4.0 placental scars per female and older females had an average of 4.8 placental scars per female. There was a 50:50 sex ratio at birth.

(a) Calculate the values of r_c and r_m for this population.

(b) Determine the transition or Leslie matrix for this population.

(c) Suppose a population with an excess of males was started with ten 1-year-old females and 20 2-year-old females just prior to breeding. Calculate the number of females in each age class over the next 3 years.